

MATHS

23.12.23

$$\begin{aligned}
 1 \quad & \sqrt{3}, 2\sqrt{3}, 3\sqrt{3}, \dots \\
 & a = \sqrt{3} \\
 & d = 2\sqrt{3} - \sqrt{3} \\
 & = \sqrt{3} \\
 & a_n = a + (n-1)d \\
 & a_{100} = \sqrt{3} + (100-1)\sqrt{3} \\
 & = \sqrt{3} + 99\sqrt{3} \\
 & = 100\sqrt{3}
 \end{aligned}$$

2 A circle can have an infinite number of tangents

$$\begin{aligned}
 3 \quad & n^{\text{th}} \text{-term of AP} = 8n - 5 \\
 & a = 8(1) - 5 \\
 & = 3
 \end{aligned}$$

Last term is 28th term

$$\begin{aligned}
 l &= 8(28) - 5 \\
 &= 219
 \end{aligned}$$

$$S_n = \frac{n}{2}(a+l)$$

$$\begin{aligned}
 S_{28} &= \frac{28}{2}(3+219) \\
 &= 3108
 \end{aligned}$$

4 $OP = OQ$ (Radii of same circle)
 $\therefore \triangle POQ$ is an isosceles triangle
 $\angle OPA = \angle OQP$ (Equal sides to equal angles)

$$OP \perp PT$$

$$\angle OPT = 90^\circ$$

$$\angle OPQ + \angle QPT = 90^\circ$$

$$55^\circ + \angle TPQ = 90^\circ$$

$$\angle TPQ = 90^\circ - 55^\circ$$
$$= 35^\circ$$

$$AP: 3, 15, 27, 39, \dots$$

$$a = 3 \quad d = 15 - 3$$
$$= 12$$

$$a_n = a + (n-1)d$$

$$n = 54$$

$$a_{54} = 3 + (54-1)12$$
$$= 639$$

Terms which is 132 more than 54th term is

$$639 + 132 = 771$$

$$a_n = 771$$

$$771 = a + (n-1)d$$

$$771 = 3 + (n-1)12$$

$$n = 65$$

$$PR = 12 \text{ cm}, OP = 5 \text{ cm and } OQ = 9 \text{ cm}$$

$$OP = OQ \perp OP$$

$$OP = 4 + 5$$

$$= 9 \text{ cm}$$

$$PO^2 = PR^2 + OP^2$$

$$PO^2 = 12^2 + 9^2$$

$$PO = 15 \text{ cm}$$

$$RO^2 = RO^2 - OO^2$$

$$RO^2 = 15^2 - 4^2$$

$$RO = \sqrt{209} \text{ cm}$$

$$OP = 5 \text{ cm}$$

$$RO^2 = PR^2 + OP^2$$

$$RO^2 = 15^2$$

$$RO = 12 \text{ cm}$$

$$PO^2 = RO^2 - OO^2$$

$$RO^2 = 13^2 - 4^2$$

$$RO = \sqrt{153} \text{ cm}$$

7

$$\text{Lowest 3 digit number} = 87 \times 2$$

$$= 174$$

$$\text{Highest 3 digit number} = 87 \times 11$$

$$= 957$$

$$d = a + (n-1)d$$

$$957 = 174 + (n-1) 87$$

$$n = 10$$

$$OP = 10 \text{ cm}$$

$$OT = 8 \text{ cm}$$

$$OT \perp PT$$

$$\text{Radius} =$$

$$\text{In } \triangle O$$

$$\text{Using}$$

$$OP$$

$$10^2$$

$$PT$$

$$P$$

$$P$$

$$\text{Long}$$

$$9$$

$$a_2$$

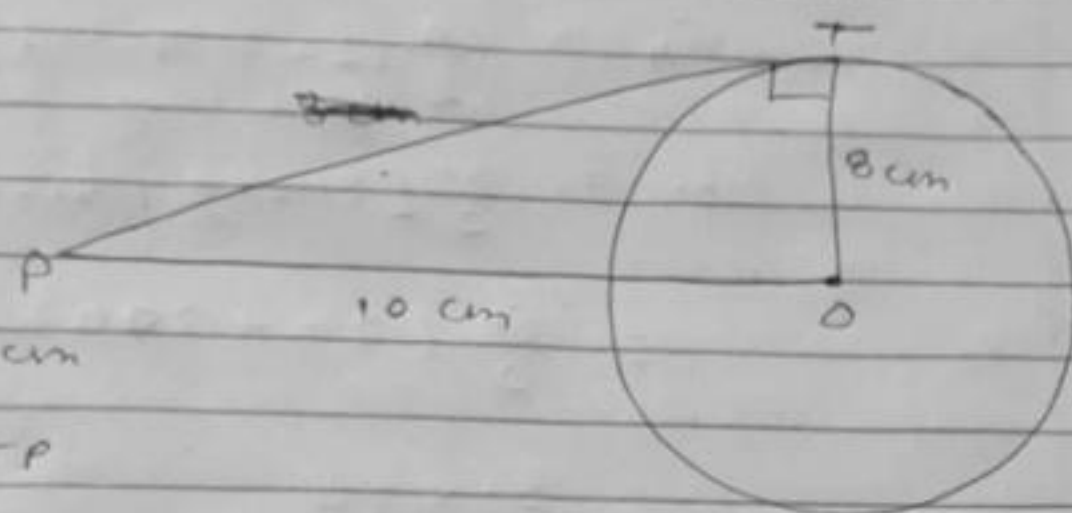
$$a + d$$

$$OP = 10 \text{ cm}$$

$$OT = 8 \text{ cm}$$

$$OT \perp PT$$

$$\text{Radius} = 8 \text{ cm}$$



In $\triangle OTP$

Using Pythagoras Theorem

$$OP^2 = OT^2 + PT^2$$

$$10^2 = 8^2 + PT^2$$

$$PT^2 = 10^2 - 8^2$$

$$PT^2 = 100 - 64$$

$$= 36$$

$$PT = 6 \text{ cm}$$

Length of tangent = 6 cm

$$a_2 = 14$$

$$a_3 = 18$$

$$a + d = 14 \quad \text{--- (1)}$$

$$a + 2d = 18 \quad \text{--- (2)}$$

$$a + d = 14$$

$$a + 2d = 18$$

$$-d = -4$$

$$\boxed{d = 4}$$

Putting value of d in eq (1)

$$a + 4 = 14$$

$$\boxed{a = 10}$$

$$S_n = \frac{n}{2} [2a + (n-1)d]$$

$$S_{51} = \frac{51}{2} [2(10) + (51-1)d]$$

$$= \frac{51}{2} (20 + 200)$$

$$= 51 \times 110$$

$$S_{51} = 5610$$

10 ABCD is a parallelogram circumscribed in circle

$$AB = CD \quad \text{--- (1)}$$

$$BC = AD \quad \text{--- (2)}$$

$$\left. \begin{array}{l} DR = DS \\ CR = CQ \\ BP = BQ \\ AP = AS \end{array} \right\} \text{(Tangents on the circle from the point are same)}$$

Adding all these equations we get

$$DR + CR + BP + AP = DS + CQ + BQ + AS$$

$$CD + AB = AD + BC$$

Putting the value of eq (1) and (2)

$$2AB = 2BC$$

$$AB = BC \quad \text{--- (3)}$$

From eq (1) (2) and (3) we get

$$AB = BC = CD = DA$$

$\therefore$ ABCD is a Rhombus

